

WEATHER DATA PREPROCESSING AND CLIMATE ANALYSIS REPORT

1 Introduction

Weather data analysis plays a crucial role in understanding climate patterns, environmental conditions, and atmospheric behavior across different geographical regions. With the increasing availability of global weather datasets, data preprocessing and visualization techniques can be applied to extract meaningful insights.

This project focuses on analyzing a Global Weather Dataset containing over 126,000 records and 41 weather-related parameters. The dataset includes temperature, humidity, wind speed, pressure, visibility, air quality indicators, and astronomical information such as sunrise and sunset.

The main goal of this project is to:

- Preprocess the dataset
- Identify seasonal patterns
- Analyze temperature variations
- Study relationships between weather parameters
- Extract insights using data visualization

2 Dataset Description

The dataset contains weather observations collected from multiple countries worldwide.

Key Attributes Include:

- Temperature (Celsius & Fahrenheit)
- Feels-like Temperature
- Humidity

- Wind Speed (mph & kph)
- Atmospheric Pressure
- Precipitation
- Visibility
- Air Quality Index (US EPA & UK DEFRA)
- Pollutant Levels (PM2.5, PM10, CO, NO₂, O₃, SO₂)
- Sunrise & Sunset Times

Dataset Size:

- Records: 126,000+
- Columns: 41
- Data Type: Structured CSV

The dataset provides both meteorological and environmental indicators, making it suitable for comprehensive climate analysis.

3 Data Preprocessing

Data preprocessing is an essential step before performing any analysis.

The following steps were performed:

3.1 Data Inspection

- Used `df.info()` to check data types
- Used `df.isnull().sum()` to identify missing values
- Used `df.describe()` for statistical summary

Observation:

- No missing values were detected.
- All numerical columns were properly formatted.

3.2 Duplicate Removal

Duplicate records were checked and removed using:

```
df.drop_duplicates(inplace=True)
```

This ensures data integrity and prevents bias.

3.3 Date Conversion

The last_updated column was converted into datetime format:

```
df["last_updated"] = pd.to_datetime(df["last_updated"])
```

This allowed extraction of:

```
df["Month"] = df["last_updated"].dt.month
```

3.4 Season Classification

Seasons were assigned based on month:

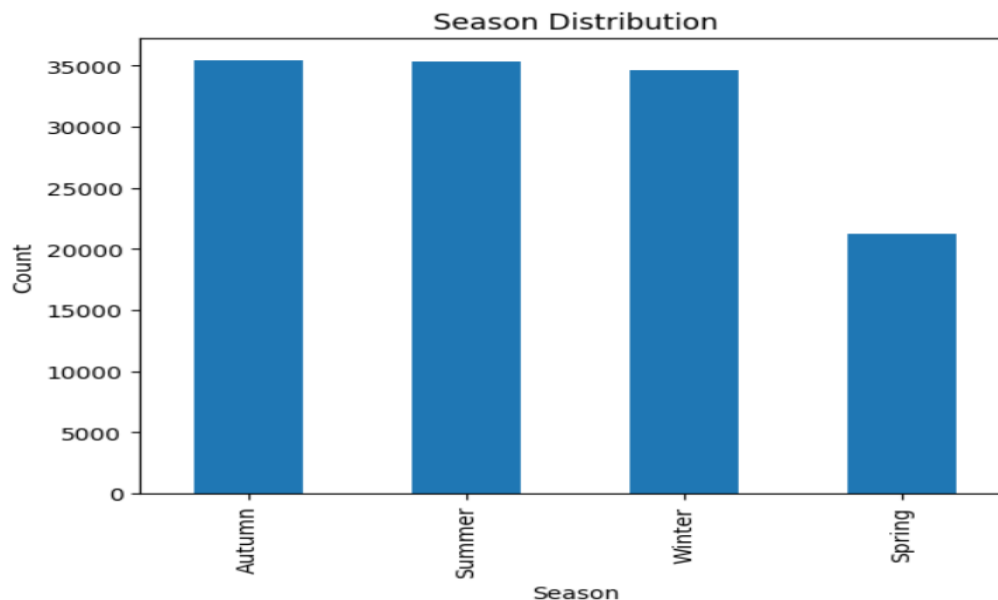
- Winter → Dec, Jan, Feb
- Spring → Mar, Apr, May
- Summer → Jun, Jul, Aug
- Autumn → Sep, Oct, Nov

```
df["Season"] = df["Month"].apply(get_season)
```

This classification enabled seasonal analysis of climate data.

Data Visualization and Analysis

4.1 Season Distribution



(Season Distribution Bar Chart)

Objective:

To analyze how weather records are distributed across seasons.

Observation:

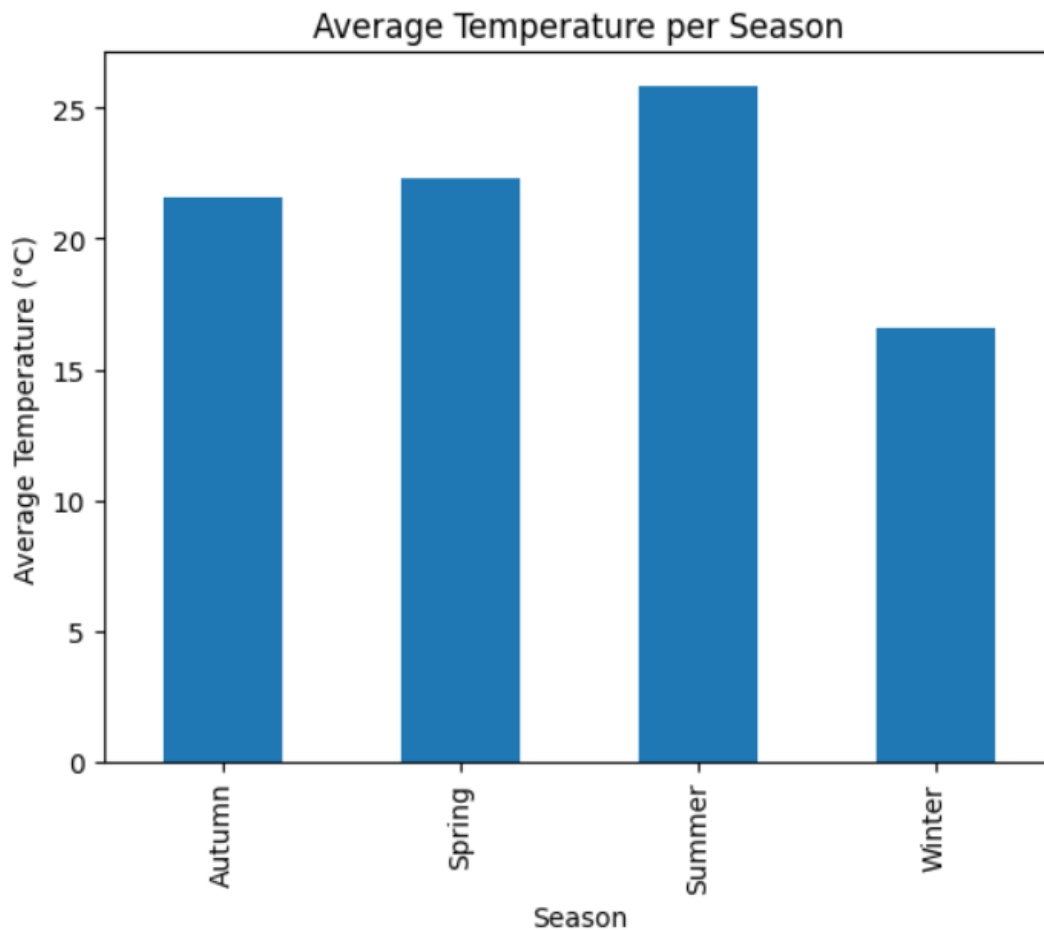
- The dataset is not evenly distributed.
- One season contains significantly more observations.

Insight:

The dataset likely represents data collected within a limited time frame.

Conclusion:

Seasonal imbalance may influence comparative seasonal studies.

4.2 Average Temperature per Season

(Average Temperature per Season Bar Chart)

Objective:

To compare temperature variations across seasons.

Observation:

- Summer has the highest average temperature.
- Winter has the lowest average temperature.
- Clear variation exists between seasons.

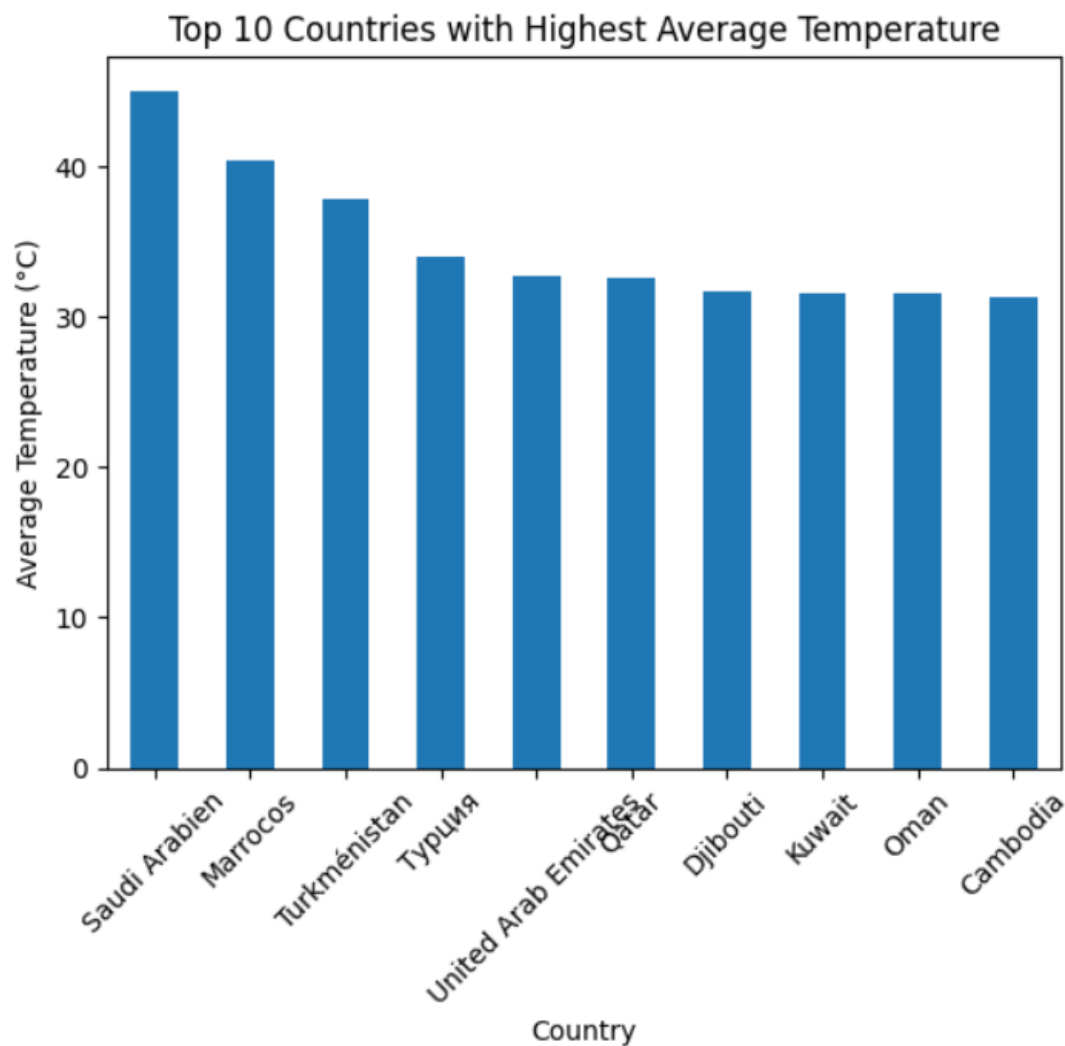
Insight:

The seasonal mapping logic is validated.

Conclusion:

Temperature patterns align with expected climatic behavior.

4.3 Top 10 Hottest Countries



(Top 10 Hottest Countries Bar Chart)

Objective:

To identify countries with the highest average temperature.

Observation:

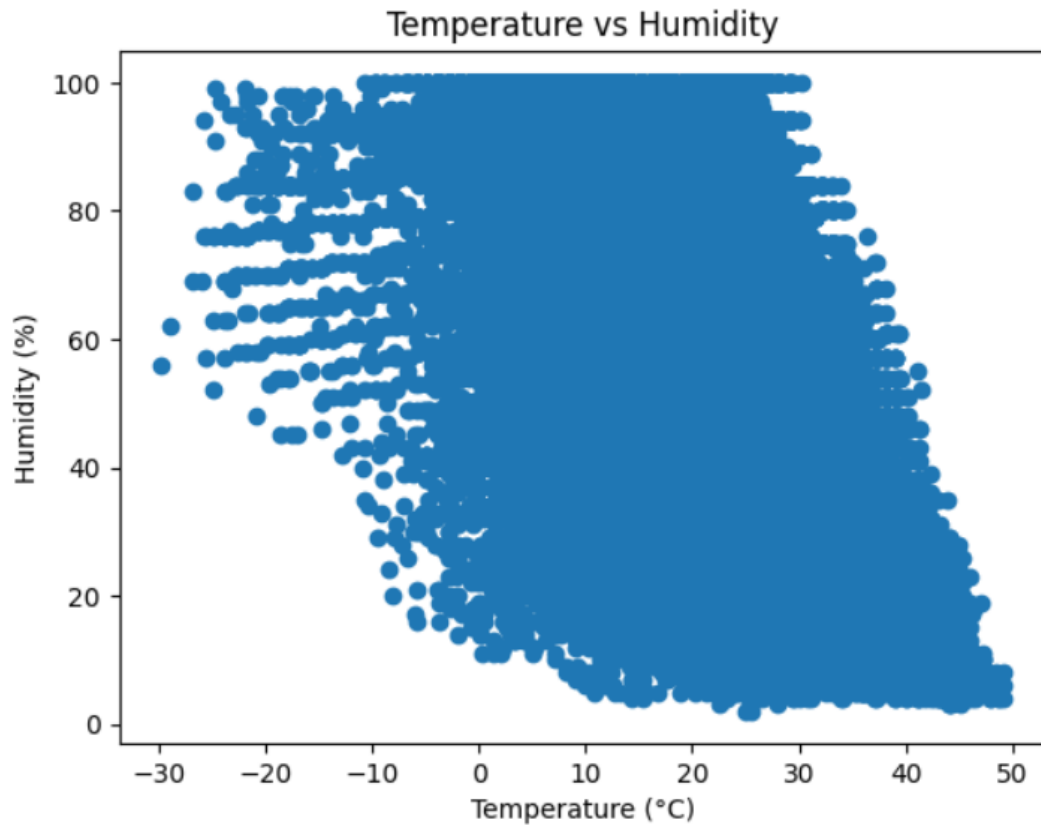
- Some countries consistently record higher temperatures.
- These countries are likely located near equatorial or desert regions.

Insight:

Geographical position significantly influences temperature.

Conclusion:

There is considerable global climate diversity.

4.4 Temperature vs Humidity Analysis**Objective:**

To study the relationship between atmospheric heat and moisture.

Observation:

- A slight negative trend is observed.

- As temperature increases, humidity tends to decrease.

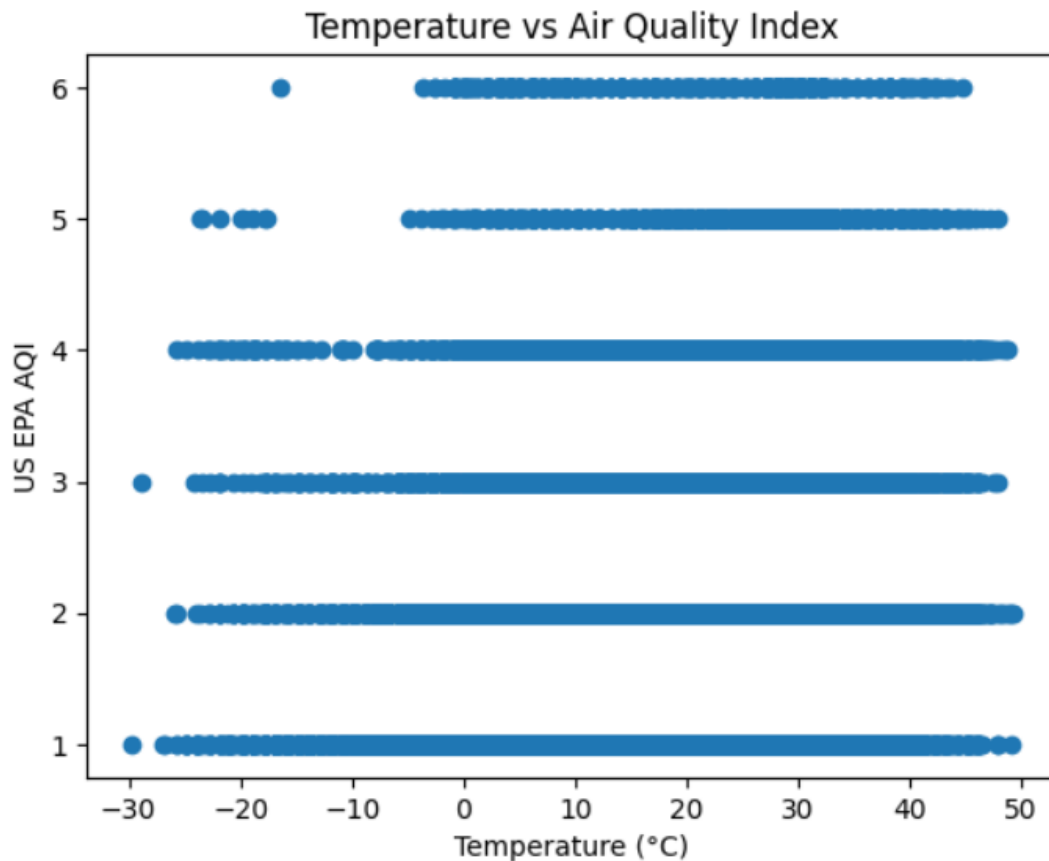
Insight:

Temperature influences moisture retention capacity of air.

Conclusion:

Temperature and humidity are moderately correlated atmospheric variables.

4.5 Temperature vs Air Quality Index



Objective:

To examine whether temperature impacts pollution levels.

Observation:

- Weak to moderate relationship observed.
- AQI does not depend solely on temperature.

Insight:

Air pollution depends on multiple factors:

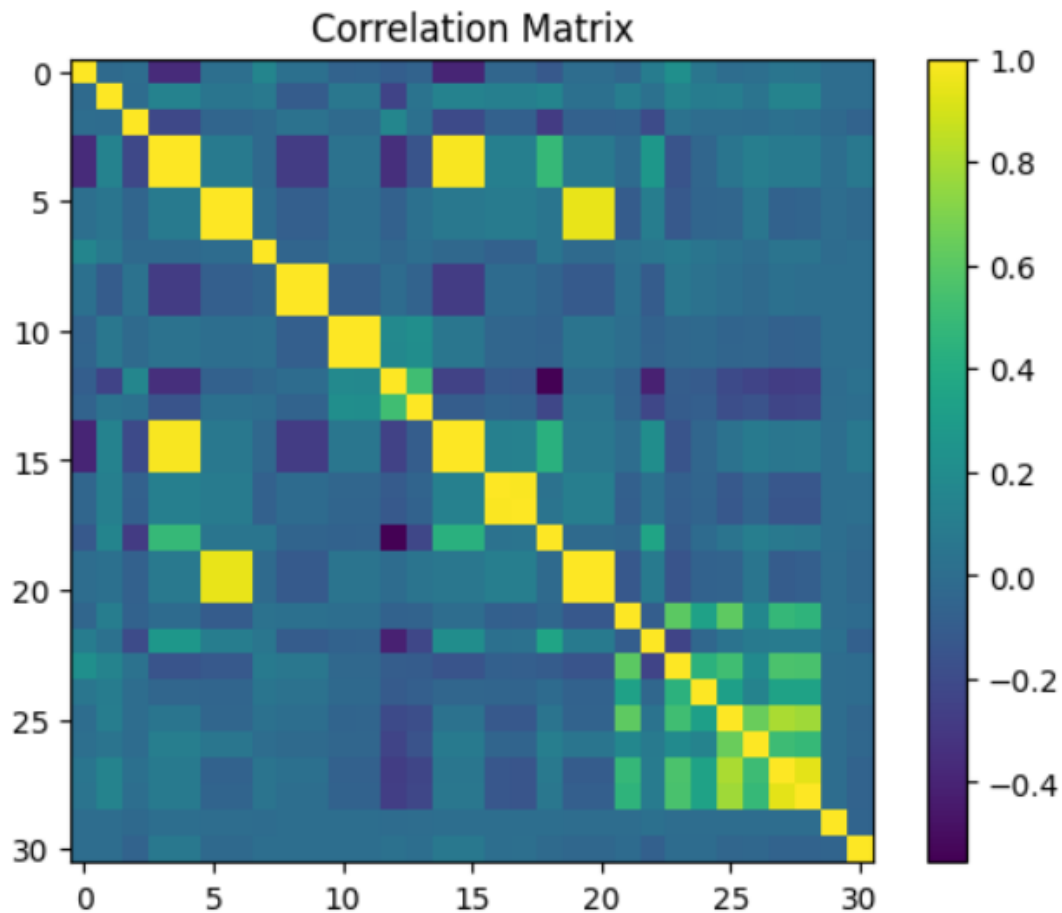
- Industrial activity

- Traffic emissions
- Wind conditions
- Geography

Conclusion:

Temperature alone cannot predict air quality levels.

4.6 Correlation Matrix



Objective:

To identify relationships among numerical weather parameters.

Key Findings:

- Strong positive correlation between:
 - Temperature and Feels-like Temperature
 - Wind mph and Wind kph

- Visibility km and Visibility miles
- Moderate correlation among air quality pollutants.

Insight:

Certain weather parameters are interdependent.

Conclusion:

Correlation analysis helps identify influential variables for predictive modeling.
